

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2-Industry Problem Solving

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4– Industry Problem Solving	Assessment:	Assessment Tool 2-Industry Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO4 Statement:	Excellent understanding and accurate application of population, sample, parameters, statistics, and point estimation.. (BL4)		
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Section / Batch:	B. Tech AIML	Date:	19-08-2026

Code of Conduct

I K. lahari (192325118) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: K. Lahari

Criteria	Weight age	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Hypothesis Testing Using Confidence Interval and P-Value Approach

1. Digital Marketing — Hypothesis Testing Using CI

Before introducing a new advertisement, the conversion rate was 8%. After the campaign, 90 out of 1,000 visitors converted. Using a 95% confidence interval, determine whether the observed conversion rate provides evidence of improvement.

Solution

1. Population and Sample: Population: all website visitors exposed to the advertisement campaign.

Sample: $n = 1,000$ visitors who viewed the new advertisement.

2. Parameter of Interest: The true population proportion of visitors who convert after the campaign, denoted p .

3. Point Estimate: $\hat{p} = x/n = 90/1000 = 0.09$ (9%).

4. Null Hypothesis (H_0): $p = 0.08$ (the campaign has not improved the conversion rate beyond the baseline of 8%).

5. Alternative Hypothesis (H_1): $p > 0.08$ (the campaign has improved the conversion rate).

6. Significance Level (α): $\alpha = 0.05$ (95% confidence).

7. Confidence Interval / Test Statistic: Standard error: $SE = \sqrt{[\hat{p} (1-\hat{p})/n]} = \sqrt{[(0.09 \times 0.91)/1000]} =$

$\sqrt{0.0000819} \approx 0.00905$. 95% CI = $\hat{p} \pm 1.96 \times SE = 0.09 \pm 1.96(0.00905) = 0.09 \pm 0.0177 = (0.0723, 0.1077)$, i.e. (7.23%, 10.77%).

8. P-value (equivalent check): $z = (0.09 - 0.08) / \sqrt{[(0.08 \times 0.92) / 1000]} = 0.01 / 0.00858 \approx 1.17$; onetailed p-value = $P(Z > 1.17) \approx 0.121$.

9. Compare with α : The baseline value 0.08 lies inside the 95% CI (0.0723, 0.1077), and the pvalue (0.121) is greater than α (0.05).

10. Industry-Oriented Conclusion: There is not enough statistical evidence at the 95% confidence level to conclude that the new advertisement has improved the conversion rate. Although the observed rate (9%) is numerically higher than the baseline (8%), the increase could plausibly be due to random sampling variation. The marketing team should continue testing with a larger sample or run the campaign longer before rolling it out broadly.

2. Education Technology — Hypothesis Testing Using CI

An EdTech company claims that its adaptive-learning platform improves the average test score to at least 75. A sample of 64 students obtains an average score of 78 with standard deviation. Use a 95% confidence interval to assess the claim.

Solution

1. Population and Sample: Population: all students using the adaptive-learning platform.

Sample: $n = 64$ students.

2. Parameter of Interest: The true population mean test score, μ , of students using the platform.

3. Point Estimate: $\bar{x} = 78$, with sample standard deviation $s = 10$.

4. Null Hypothesis (H_0): $\mu = 75$ (the platform raises the average score to exactly the claimed minimum of 75).

5. Alternative Hypothesis (H_1): $\mu > 75$ (the platform's true average score exceeds 75, supporting or exceeding the claim).

6. Significance Level (α): $\alpha = 0.05$ (95% confidence).

7. Confidence Interval / Test Statistic: Standard error: $SE = s/\sqrt{n} = 10/\sqrt{64} = 10/8 = 1.25$. 95% CI = $\bar{x} \pm 1.96 \times SE = 78 \pm 1.96(1.25) = 78 \pm 2.45 = (75.55, 80.45)$.

8. P-value (equivalent check): $z = (78 - 75)/1.25 = 2.40$; one-tailed p-value = $P(Z > 2.40) \approx 0.0082$.

9. Compare with α : The claimed value of 75 lies below the entire 95% CI (75.55, 80.45), and the p-value (0.0082) is less than α (0.05).

10. Industry-Oriented Conclusion: There is sufficient statistical evidence at the 95% confidence level to support the EdTech company's claim — the platform improves average test scores to at least 75, and the true mean is likely even higher (between roughly 75.55 and 80.45).

This CI strengthens the company's marketing claim and can be cited in product validation reports.

3. Automobile Industry — Hypothesis Testing Using P-value

A car manufacturer claims that its new model provides an average mileage of 20 km/litre. A sample of 50 cars gives a mean mileage of 19.2 km/litre with standard deviation 2.5 km/litre. Perform a hypothesis test at the 5% significance level using the p-value approach.

Solution

- 1. Population and Sample:** Population: all cars of the new model produced by the manufacturer.

Sample: $n = 50$ cars tested for mileage.

- 2. Parameter of Interest:** The true population mean mileage, μ , of the new car model (km/litre).

3. **Point Estimate:** $\bar{x} = 19.2$ km/litre, with sample standard deviation $s = 2.5$ km/litre.
4. **Null Hypothesis (H0):** $\mu = 20$ km/litre (the manufacturer's claim is true).
5. **Alternative Hypothesis (H1):** $\mu \neq 20$ km/litre (the true average mileage differs from the claimed value — two-tailed test).
6. **Significance Level (α):** $\alpha = 0.05$.
7. **Test Statistic:** Standard error: $SE = s/\sqrt{n} = 2.5/\sqrt{50} = 2.5/7.071 \approx 0.3536$. $z = (\bar{x} - \mu_0)/SE = (19.2 - 20)/0.3536 = -0.8/0.3536 \approx -2.263$.
8. **P-value:** Two-tailed p-value $= 2 \times P(Z < -2.263) \approx 2 \times 0.0118 = 0.0236$.
9. **Compare with α :** p-value (0.0236) $< \alpha$ (0.05), so the result is statistically significant.
10. **Industry-Oriented Conclusion:** There is sufficient evidence to reject H0.
The new model's actual average mileage (19.2 km/litre) is significantly lower than the claimed 20 km/litre. The manufacturer should review the mileage claim used in marketing and specifications, as it may not hold up under quality-testing/regulatory scrutiny, or should investigate manufacturing/design factors causing the shortfall.

4. Cloud Computing — Hypothesis Testing Using P-value

A cloud provider claims that its average server response time is 100 ms. A sample of 81 requests has a mean of 104 ms and standard deviation of 12 ms. Test the claim at $\alpha = 0.05$ using the p value method.

Solution

1. **Population and Sample:** Population: all server requests handled by the cloud provider.

Sample: $n = 81$ requests.

2. **Parameter of Interest:** The true population mean server response time, μ (in milliseconds).
3. **Point Estimate:** $\bar{x} = 104$ ms, with sample standard deviation $s = 12$ ms.
4. **Null Hypothesis (H0):** $\mu = 100$ ms (the provider's claimed average response time is correct).
5. **Alternative Hypothesis (H1):** $\mu \neq 100$ ms (the true average response time differs from the claim — two-tailed test).
6. **Significance Level (α):** $\alpha = 0.05$.
7. **Test Statistic:** Standard error: $SE = s/\sqrt{n} = 12/\sqrt{81} = 12/9 = 1.333$. $z = (\bar{x} - \mu_0)/SE = (104 - 100)/1.333 \approx 3.00$.
8. **P-value:** Two-tailed $p\text{-value} = 2 \times P(Z > 3.00) \approx 2 \times 0.00135 = 0.0027$.
9. **Compare with α :** $p\text{-value} (0.0027) < \alpha (0.05)$, so the result is statistically significant.
10. **Industry-Oriented Conclusion:** There is sufficient evidence to reject H_0 .
The server's actual average response time (104 ms) is significantly higher than the claimed 100 ms. This indicates a real performance gap, not random variation. The cloud provider should investigate infrastructure bottlenecks (network latency, server load, or scaling issues) and consider revising its published SLA (Service Level Agreement) response-time commitment.

5. E-commerce — Hypothesis Testing Using P-value

An online company claims that its product return rate is 10%. In a sample of 1,000 orders, 130 products were returned. Test whether the actual return rate is significantly higher than 10% using the p-value approach at the 5% significance level.

Solution

1. **Population and Sample:** Population: all orders placed with the online company. Sample: $n = 1,000$ orders.
2. **Parameter of Interest:** The true population proportion of orders returned, p .
3. **Point Estimate:** $\hat{p} = x/n = 130/1000 = 0.13$ (13%).
4. **Null Hypothesis (H0):** $p = 0.10$ (the company's claimed return rate of 10% is correct).
5. **Alternative Hypothesis (H1):** $p > 0.10$ (the actual return rate is higher than claimed — onetailed test).
6. **Significance Level (α):** $\alpha = 0.05$.
7. **Test Statistic:** Standard error under H0: $SE = \sqrt{p_0(1-p_0)/n} = \sqrt{(0.10 \times 0.90)/1000} = \sqrt{0.00009} \approx 0.009487$. $z = (\hat{p} - p_0)/SE = (0.13 - 0.10)/0.009487 \approx 3.162$.
8. **P-value:** One-tailed p-value = $P(Z > 3.162) \approx 0.00078$.
9. **Compare with α :** p-value (0.00078) $< \alpha$ (0.05), so the result is statistically significant.
10. **Industry-Oriented Conclusion:** There is sufficient evidence to reject H0. The actual product return rate (13%) is significantly higher than the claimed 10%. This suggests underlying issues such as product quality, inaccurate listings, or sizing/fit problems. The e-commerce company should investigate root causes of returns and update its reported return-rate metric for stakeholders and investors.

Q1. A university claims that at least 80% of its students are satisfied with online classes. A survey produces a 95% confidence interval of (74%, 79%) for the satisfaction proportion. What conclusion can be made about the university's claim?

Given:

- University claims that at least 80% of students are satisfied.
- 95% Confidence Interval = (74%, 79%)

Hypotheses:

- $H_0: p \geq 0.80$
- $H_1: p < 0.80$

Analysis:

The 95% confidence interval for the satisfaction proportion is from 74% to 79%. The claimed value of 80% does not fall within this interval. In fact, the entire interval is below 80%.

Conclusion:

Therefore, the university's claim that at least 80% of students are satisfied with online classes is not supported by the survey data. The results indicate that the actual satisfaction proportion is likely to be less than 80%.

Q2. A mobile application developer claims that users spend an average of 30 minutes per day on the application. A sample produces a 99% confidence interval of (31.5, 35.2) minutes. Using the confidence interval, determine whether the claim is supported.

Given:

- Developer claims average usage = 30 minutes/day
- 99% Confidence Interval = (31.5, 35.2) minutes

Hypotheses:

- $H_0: \mu = 30$ minutes
- $H_1: \mu \neq 30$ minutes

Analysis:

The 99% confidence interval ranges from 31.5 to 35.2 minutes. The claimed average of 30 minutes is outside the confidence interval. Since the complete interval is greater than 30 minutes, the sample provides evidence that the actual average usage is higher than the claimed value.

Conclusion:

Therefore, the developer's claim that users spend an average of 30 minutes per day is not supported. The data suggests that users spend approximately 31.5 to 35.2 minutes per day on average.

Q3. A credit-card company claims that its fraud detection system identifies 95% of fraudulent transactions. After testing the system on a sample, the calculated p-value is 0.02. At a 5% significance level, what conclusion should the data scientist make?

Given:

- Company claims fraud detection rate = 95%
- p-value = 0.02
- Significance level $\alpha = 0.05$

Hypotheses:

- $H_0: p = 0.95$
- $H_1: p \neq 0.95$

Decision Rule:

If: $p\text{-value} < \alpha \rightarrow \text{Reject } H_0$

Here:

$$0.02 < 0.05$$

Therefore, we reject H_0 .

Conclusion:

Since the p-value is less than the 5% significance level, there is statistically significant evidence against the company's claim that the fraud detection system identifies 95% of fraudulent transactions.

Q4. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given:

- Existing conversion rate = 12%
- New conversion rate = 15%
- p-value = 0.03
- $\alpha = 0.05$

Hypotheses:

- H_0 : New algorithm does not improve conversion rate.
- H_1 : New algorithm improves conversion rate.

Decision Rule:

If:

$$p\text{-value} < \alpha \rightarrow \text{Reject } H_0$$

Here:

$$0.03 < 0.05$$

Therefore, we reject H_0 .

Analysis:

The observed conversion rate increased from 12% to 15%, which is an improvement of 3 percentage points. The p-value of 0.03 indicates that this improvement is statistically significant at the 5% significance level.

Conclusion:

Therefore, the new recommendation algorithm significantly improves the conversion rate compared with the existing algorithm.

Q5. An e-commerce company introduces a new recommendation algorithm. The existing algorithm has a conversion rate of 12%. After implementation, the observed conversion rate is 15%. The hypothesis test produces a p-value of 0.03. At $\alpha = 0.05$, determine whether the new algorithm significantly improves conversion.

Given:

- Purpose: Determine whether the new system reduces waiting time.
- p-value = 0.08
- Significance level $\alpha = 0.05$

Hypotheses:

- H_0 : The new system does not reduce average waiting time.
- H_1 : The new system reduces average waiting time.

Decision Rule:

If:

$$\begin{aligned} \text{p-value} < \alpha &\rightarrow \\ &\text{Reject } H_0 \end{aligned}$$

If: Here:

$$\begin{aligned} \text{p-value} \geq \alpha &\rightarrow \\ &\text{Fail to reject} \\ &H_0 \end{aligned}$$

$$0.08 > 0.05$$

Therefore, we fail to reject H_0 .

Conclusion:

Since the p-value is greater than the 5% significance level, there is insufficient statistical evidence to conclude that the AI-based scheduling system reduces the average waiting time.

The hospital should not claim that the new system reduces waiting time based on this test.